

LAB EXPERIMENT RECORD

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Date	1t october 2026
Day	Tuesday

SET 4 — PRODUCT INVENTORY MANAGEMENT

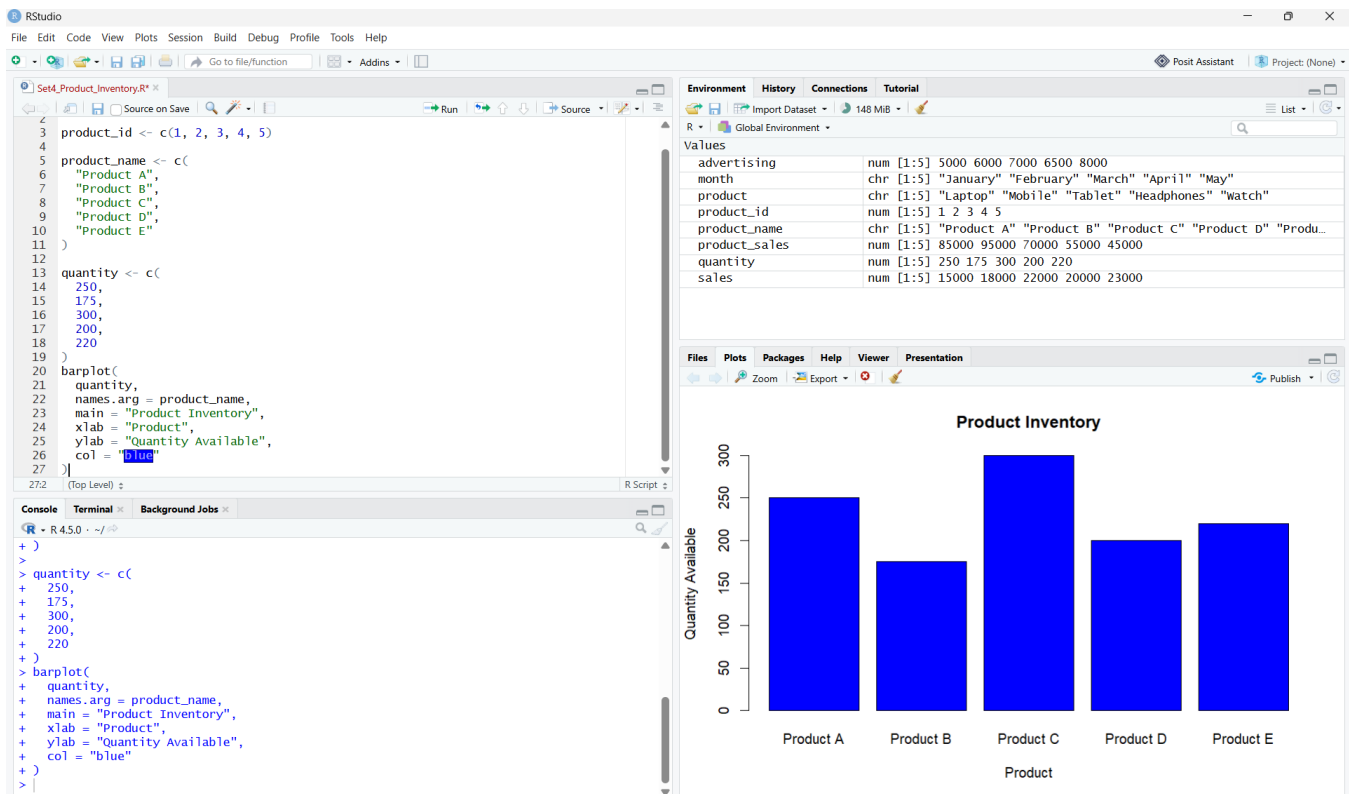
Aim

To analyze and visualize product inventory data using bar charts, stacked bar charts, and scatter plots, and to develop an interactive dashboard for exploring product inventory information.

PROGRAM 1: LINE CHART

Create a bar chart to visualize the quantity of each product.

```
barplot(  
  quantity,  
  names.arg = product_name,  
  main = "Product Inventory",  
  xlab = "Product",  
  ylab = "Quantity Available",  
  col = "blue"  
)
```



Output 1: Distribution of Customer Ages

Observation: Sales increased from January to May, with a small decrease in April.

PROGRAM 2: BAR CHART

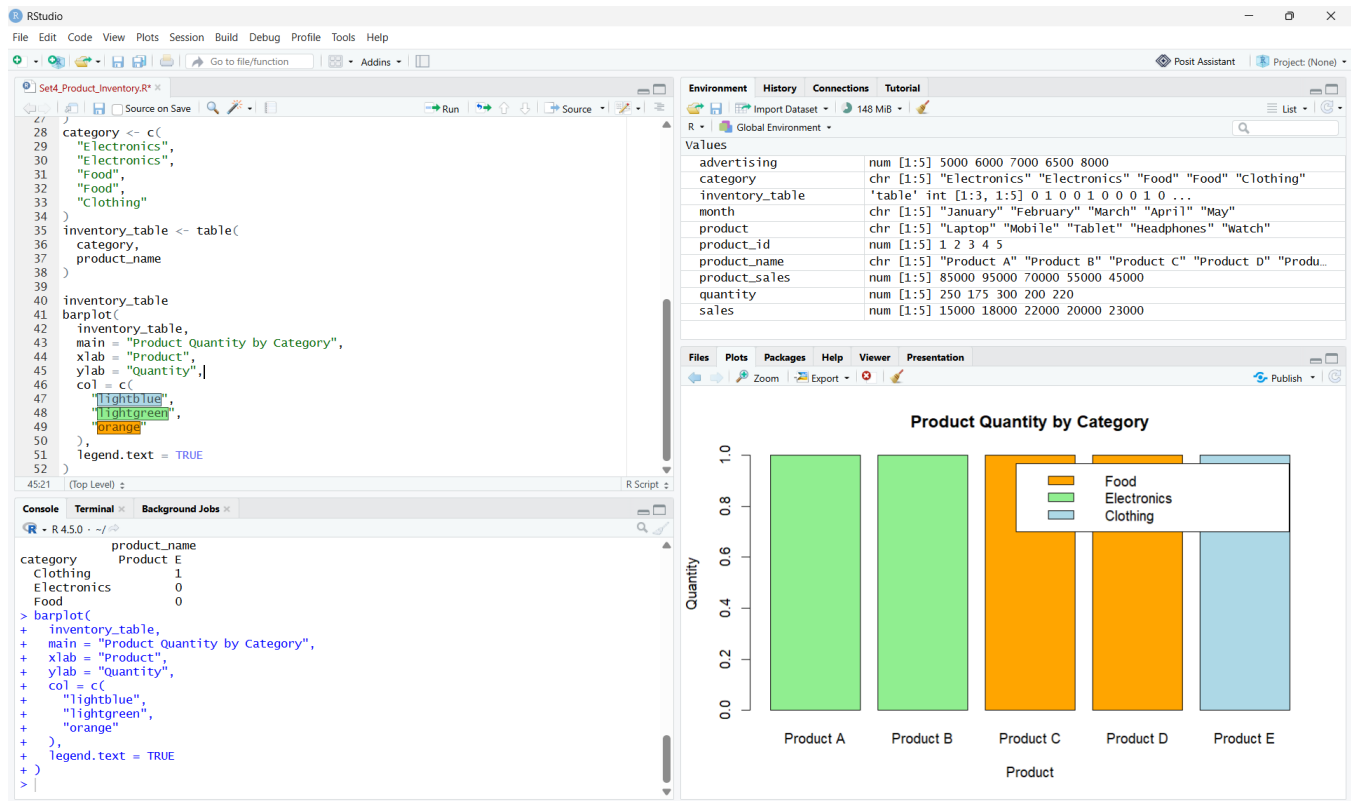
To create a stacked bar chart to visualize the quantity of products across different product categories..

```
# category <- c(
  "Electronics",
  "Electronics",
  "Food",
  "Food",
  "Clothing"
)
inventory_table <- table(
  category,
  product_name
)
inventory_table
barplot(
  inventory_table,
  main = "Product Quantity by Category",
```

```

xlab = "Product",
ylab = "Quantity",
col = c(
  "lightblue",
  "lightgreen",
  "orange"
),
legend.text = TRUE
)

```



Output 2: product quantity by category

Observation: The stacked bar chart was successfully created to represent the distribution of product quantities across different categories.

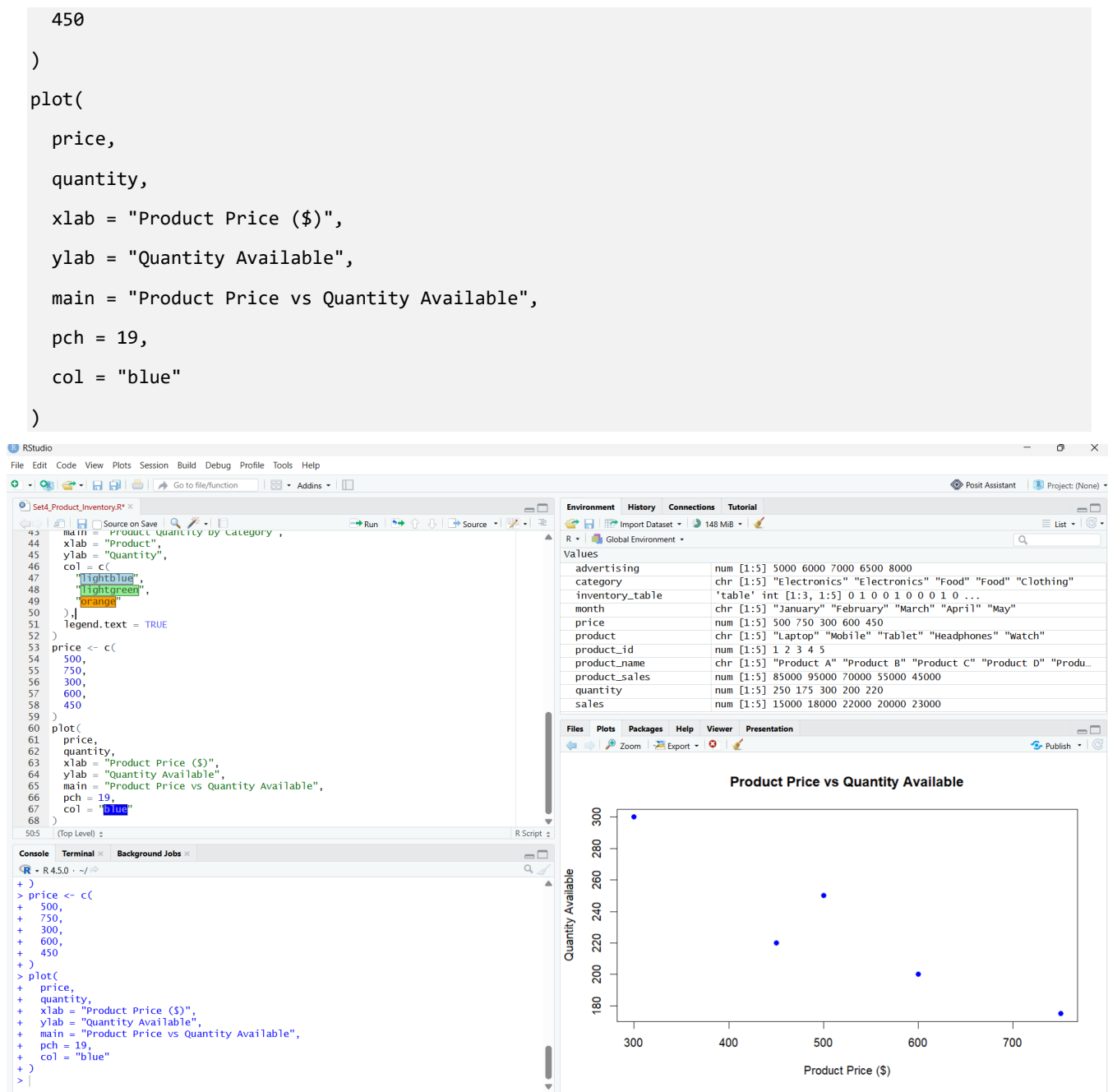
PROGRAM 3: SCATTER PLOT

To create a scatter plot to analyze the relationship between product price and quantity available.

```

# price <- c(
  500,
  750,
  300,
  600,

```



Output 3: Satisfaction Scores by Age Group

Observation: The scatter plot was successfully created to show the relationship between product price and available quantity. The graph helps identify the pattern between price and inventory level.

PROGRAM 4: Word Cloud of Customer Feedback

develop an interactive dashboard combining the product inventory bar chart and category-wise stacked bar chart.

```

library(shiny)

product_name <- c(
  "Product A",

```

```

"Product B",
"Product C",
"Product D",
"Product E"
)
quantity <- c(
  250,
  175,
  300,
  200,
  220
)
category <- c(
  "Electronics",
  "Electronics",
  "Food",
  "Food",
  "Clothing"
)
ui <- fluidPage(
  titlePanel("Product Inventory Dashboard"),
  selectInput(
    "chart",
    "Select Chart:",
    choices = c(
      "Inventory Bar Chart",
      "Category Stacked Bar Chart"
    )
  ),
  plotOutput("myplot")
)
server <- function(input, output) {
  output$myplot <- renderPlot({
    if (
      input$chart == "Inventory Bar Chart"
    ) {
      barplot(
        quantity,

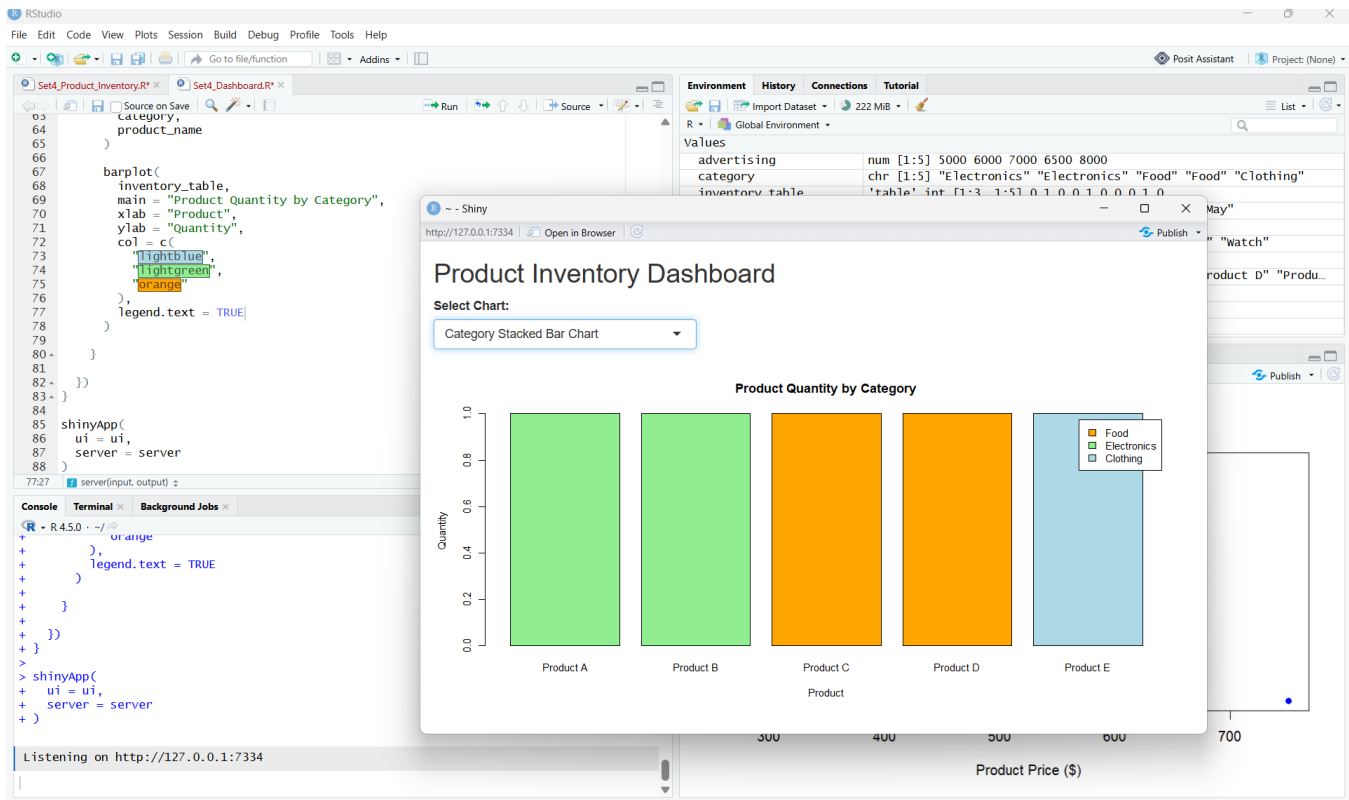
```

```

      names.arg = product_name,
      main = "Product Inventory",
      xlab = "Product",
      ylab = "Quantity Available",
      col = "blue"
    )
  } else {
    inventory_table <- table(
      category,
      product_name
    )
    barplot(
      inventory_table,
      main = "Product Quantity by Category",
      xlab = "Product",
      ylab = "Quantity",
      col = c(
        "lightblue",
        "lightgreen",
        "orange"
      ),
      legend.text = TRUE
    )
  }
})
}

shinyApp(
  ui = ui,
  server = server
)

```



Output 4: Customer Feedback Word Cloud

Observation: An interactive dashboard was successfully developed using Shiny. It allows users to switch between the product inventory and category-wise inventory charts.

Set 4 Procedure

1. Create vectors for product ID, product name, and quantity available.
2. Use `barplot()` to visualize the quantity available for each product.
3. Create product categories using `category`, form a frequency table using `table()`, and use `barplot()` to display the stacked inventory distribution.
4. Create product price data and use `plot()` to analyze the relationship between product price and quantity available.
5. Use `shiny()` to develop an interactive dashboard combining the inventory bar chart and category-wise stacked bar chart.

Set 4 Result

The product inventory data was successfully visualized using a bar chart, stacked bar chart, and scatter plot. An interactive dashboard was also developed to explore product quantities and category-wise inventory distribution.